ONLINE CASE MANAGEMENT SYSTEM (OCMS)

1. Introduction

The Online Case Management System (OCMS) is a web-based platform developed to simplify and digitize
the process of handling academic-related requests between students and staff members. In many institutions,
case handling is done manually through paperwork, which results in delays, lack of transparency, and data mismanagement.
OCMS aims to eliminate these issues by providing a centralized digital solution that ensures efficiency, transparency,
and structured record maintenance.

The system allows students to register, log in securely, and submit various academic or administrative cases
such as academic issues, library-related concerns, or hostel requests. Staff members can log in to review submitted cases
and update their status as approved or rejected along with remarks. This structured workflow ensures smooth communication
and real-time tracking of requests.

2. Problem Statement

Traditional case handling systems rely heavily on paper-based submissions. This leads to multiple problems
such as lost documents, slow approvals, lack of status visibility, and inefficient communication. Students
often remain unaware of the progress of their requests, and staff members face difficulty tracking previous records.

There is a strong need for an automated system that can securely manage submissions, provide role-based access,
and maintain a complete digital history of all cases. OCMS is designed to address these limitations.

3. Objectives

• Develop a secure and scalable web application.
• Implement role-based authentication (Student and Staff).
• Enable case submission with optional file upload.
• Provide dashboard summaries and analytics.
• Maintain processed case history records.
• Ensure data security using hashing and prepared statements.
• Create an intuitive and responsive user interface.

4. Scope of the Project

The scope of OCMS includes academic institutions where student requests require structured processing.
The system focuses on internal case handling between students and staff. It does not include third-party integrations
in Phase 1 but is designed in a way that future expansion is possible.

The application supports:
- Student Registration & Login
- Staff Login
- Case Submission
- Case Review & Decision Making
- Case History Tracking
- File Upload Support

5. Technology Stack

Frontend Technologies:
HTML5, CSS3, Bootstrap 5, JavaScript

Backend Technologies:
Core PHP running on Apache Server (XAMPP)

Database:
MySQL Relational Database

Version Control:
Git & GitHub

Security:
Password hashing using password_hash(), session-based authentication,
role-based authorization, SQL injection prevention using prepared statements,
and file type validation for uploads.

6. System Architecture

OCMS follows a Monolithic 3-Tier Architecture:

Presentation Layer – Built using HTML, CSS, and Bootstrap.
Business Logic Layer – Implemented using PHP.
Data Layer – Managed using MySQL Database.

The client sends HTTP requests to the server. The PHP backend processes the request, applies business logic, interacts with the database, and returns the response to the client.

7. Database Design

The system consists of two main tables: Users and Cases.

Users Table stores student and staff login information.
Cases Table stores all submitted cases along with their status and remarks.

Relationship:
One student can submit multiple cases (One-to-Many relationship).
The student_id in Cases table acts as a foreign key referencing the Users table.

	Table	Action	Rows	Type	Collation	Size	Overhead
☐	cases	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	notifications	Browse Structure Search Insert Empty Drop	39	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	users	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	valid_staff_ids	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
	4 tables	Sum	55	InnoDB	utf8mb4_general_ci	144.0 KiB	0 B

Check all With selected:

8. Features Implemented

- Student Registration & Login
- Staff Login
- Add Case Submission with File Upload
- Approve / Reject Cases
- Dashboard Summary Cards
- Status Tracking
- Case History Maintenance
- Secure Logout System

9. Security Implementation

Security is a critical component of OCMS.

Passwords are hashed before storing in the database.
Sessions are used for authentication management.
Role-based access ensures students cannot access staff pages.
Prepared statements prevent SQL injection.
File uploads are restricted to allowed formats such as JPG, PNG, and PDF.

10. UI/UX Design

The system uses a clean and modern dashboard layout.
The sidebar navigation allows easy access to features.
Status indicators use color coding:
Green – Approved
Red – Rejected
Blue – Pending

The interface is responsive and works across devices.

11. Testing & Validation

The application was tested for login validation, case submission accuracy,
file upload functionality, approval and rejection workflows, and database integrity.

Test cases included:
- Valid and invalid login attempts
- File upload restrictions
- Role-based access control testing
- Case status updates

12. Future Enhancements

• Email Notifications
• Admin Panel
• Advanced Search & Filtering
• Pagination Support
• Export Reports to PDF
• Dark Mode UI Theme
• Integration with institutional ERP systems

13. Project Folder Structure

```
OCMS/
 │
 ├─── index.php
 ├─── logout.php
 │
 ├─── auth/
 │    ├─── login.php
 │    ├─── register.php
 │
```

```
├─── student/
│    ├─── dashboard.php
│    ├─── add_case.php
│    ├─── my_cases.php
│    ├─── approved_cases.php
│    ├─── rejected_cases.php
│    ├─── case_history.php
│
├─── staff/
│    ├─── dashboard.php
│    ├─── received_cases.php
│    ├─── approved_cases.php
│    ├─── rejected_cases.php
│    ├─── history.php
│
├─── includes/
│    ├─── db.php
│    ├─── header.php
│    ├─── sidebar.php
│
├─── uploads/
├─── assets/
│    ├─── css/
│    ├─── js/
│    ├─── images/
```

14. Project Boilerplate Setup

The Online Case Management System repository was initialized using Git and hosted on GitHub. The project follows a well-structured folder hierarchy separating authentication, student module, staff module, shared includes, uploads, and static assets to maintain clean architecture.

15.GitHub Repository Initialization Steps:

- git init
- git add .
- git commit -m 'Initial commit - OCMS Setup'
- git branch -M main
- git remote add origin <repository-url>
- git push -u origin main

16. GitHub Workflow & Documentation

Version control is managed using Git and GitHub. The repository contains a properly configured.gitignore file, structured README documentation, and a clear branching strategy for organized development.

Initial README Documentation Includes:

Project Title: Online Case Management System (OCMS)

Project Description

Technology Stack Used

Installation Steps:

1. Clone the repository from GitHub.

2. Move the project folder to htdocs directory.

3. Import the database using phpMyAdmin.

4. Start Apache and MySQL using XAMPP.

5. Access via localhost/OCMS in browser.

Branching Strategy:

• main – Production-ready stable code.

• development – Integration branch for testing.

• feature-auth – Authentication module development.

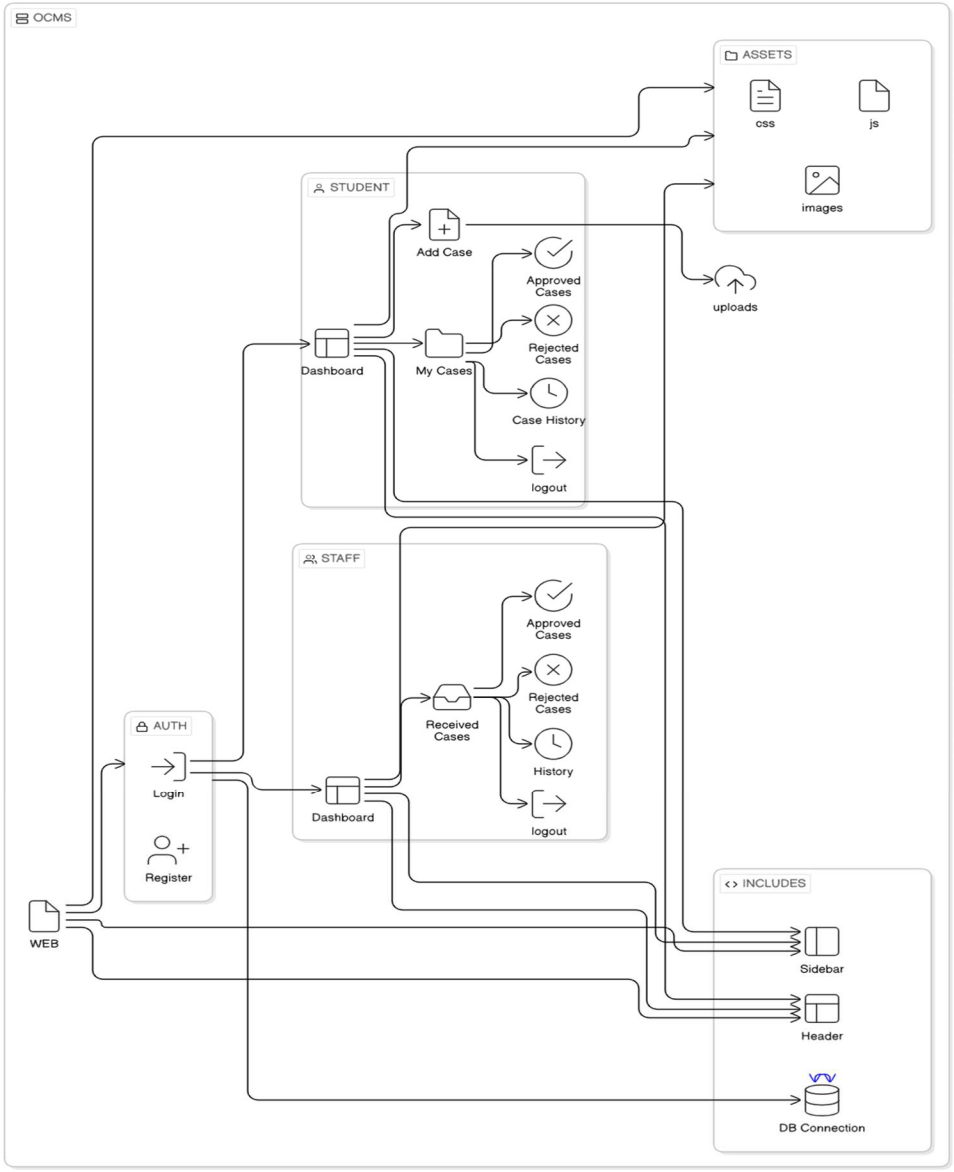
• feature-student – Student dashboard features.

• feature-staff – Staff module features.

This structured Git workflow ensures proper version tracking, feature isolation, and smooth deployment process.

14. Entity Relationship (ER) Diagram

The diagram below represents the relationship between Users and Cases tables.



15. Conclusion

The Online Case Management System successfully digitizes case handling and improves efficiency within academic institutions. It ensures structured data management,
secure authentication, and transparent communication between students and staff.

The project demonstrates practical implementation of web technologies and relational database design while maintaining security best practices.